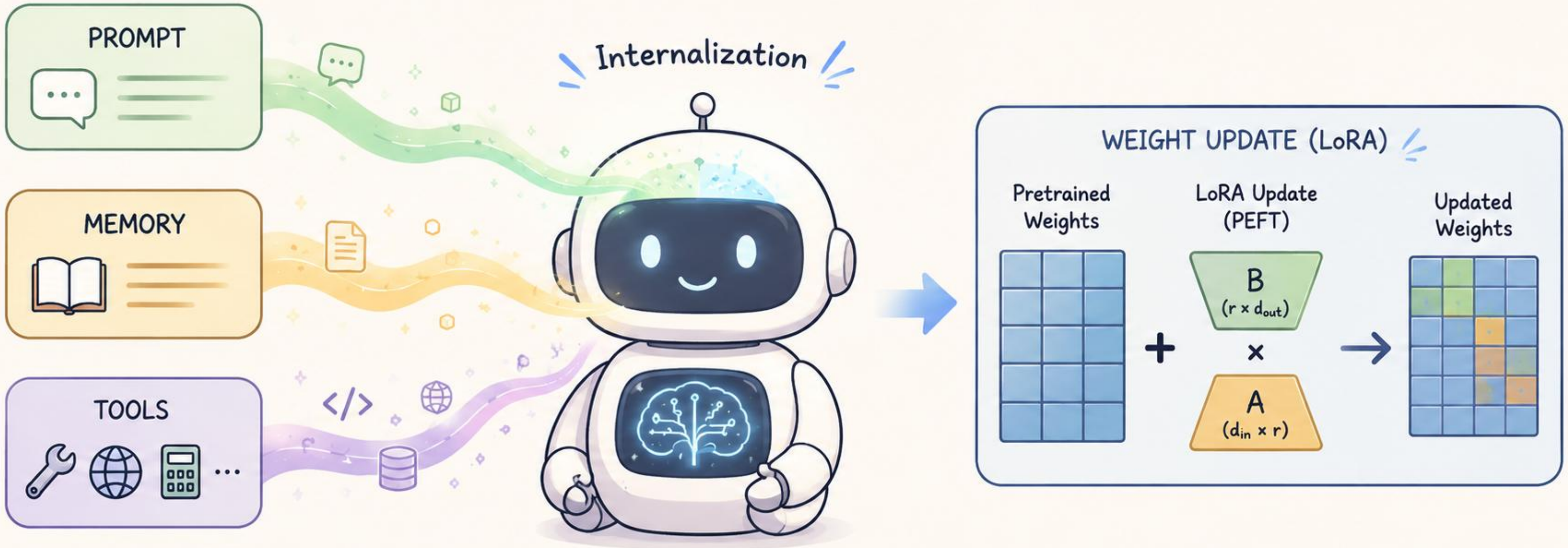
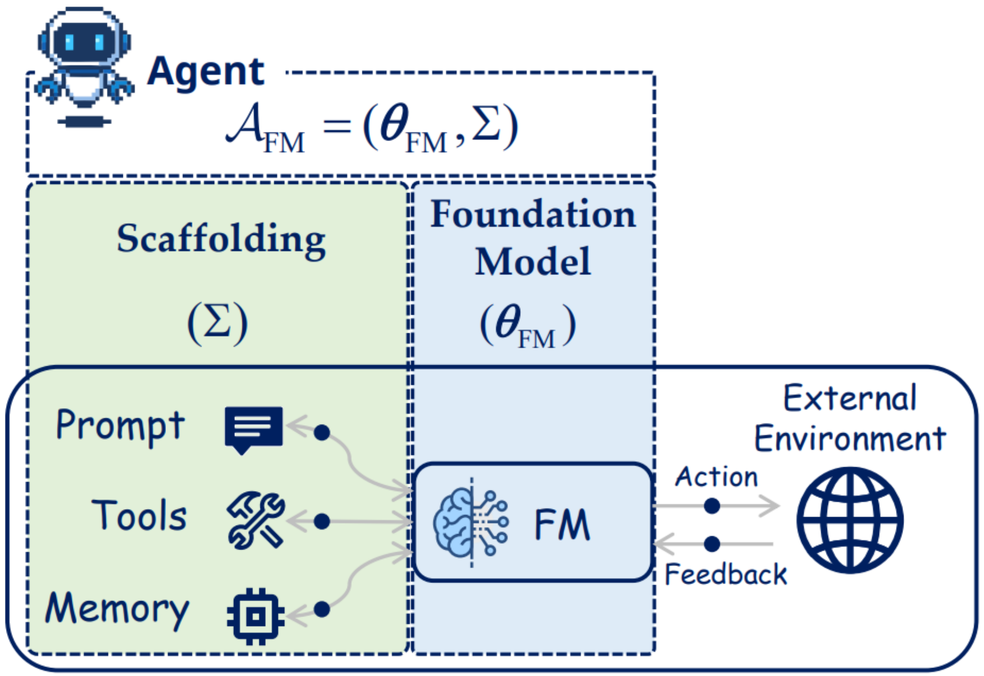
From Harness Evolution to Policy Internalization: Rethinking PEFT-RLVR for Self-Evolving Agents

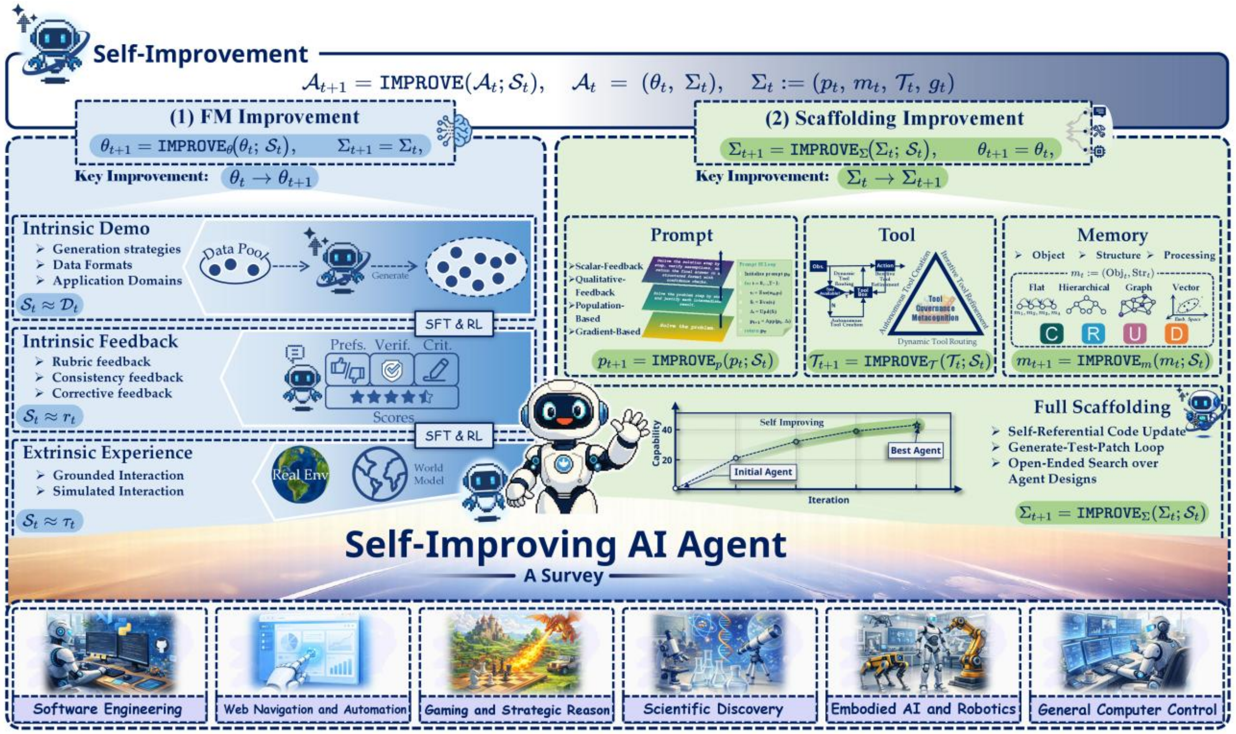


WOOJIN LEE

Survey on Self-Evolving Agents

Foundation Model-Centric Evolution vs. Scaffolding (Harness) Evolution

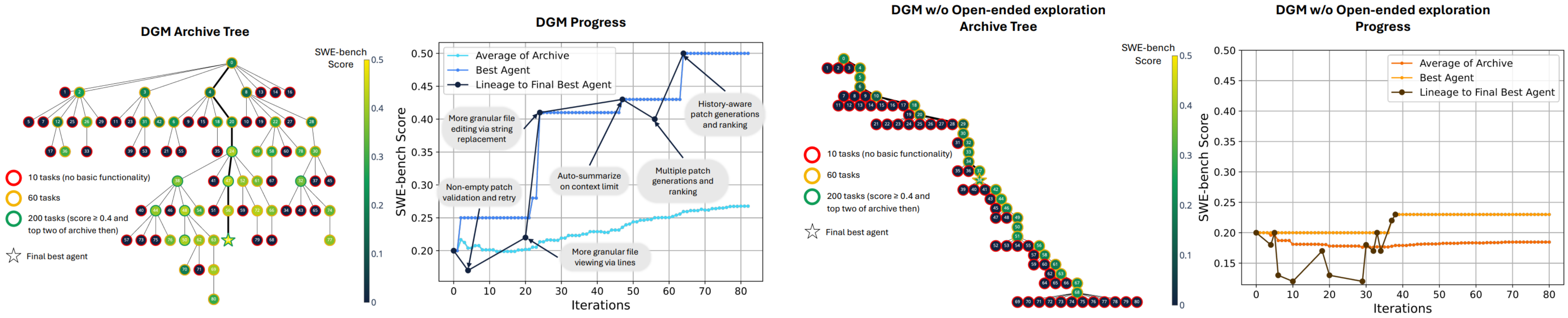




https://github.com/selfimproving-agent/Awesome-Self-Improving-Agents

Survey on Self-Evolving Agents

Focusing on Harness Evolution: DGM, GEPA, etc.

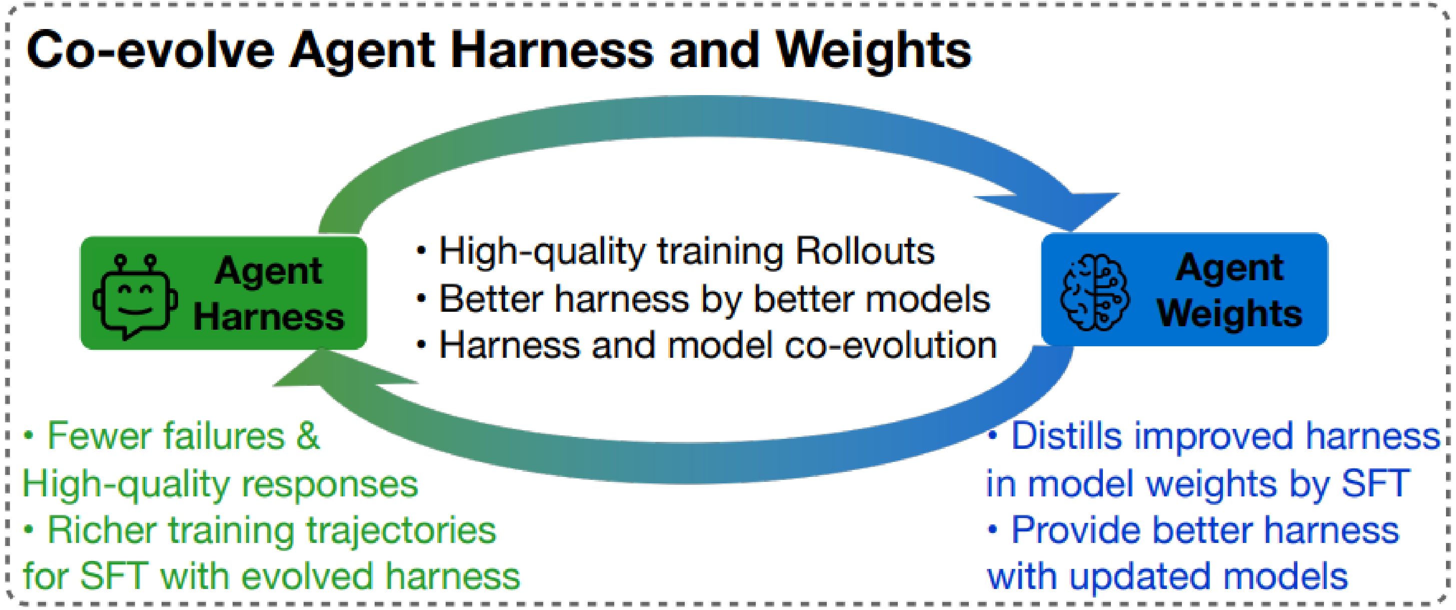


Non-selected (not best) harness can serve as stepping stones
that result in improvements much later than their original greedy discovery.

Darwin Godel Machine: Open-Ended Evolution of Self-Improving Agents: https://arxiv.org/abs/2505.22954

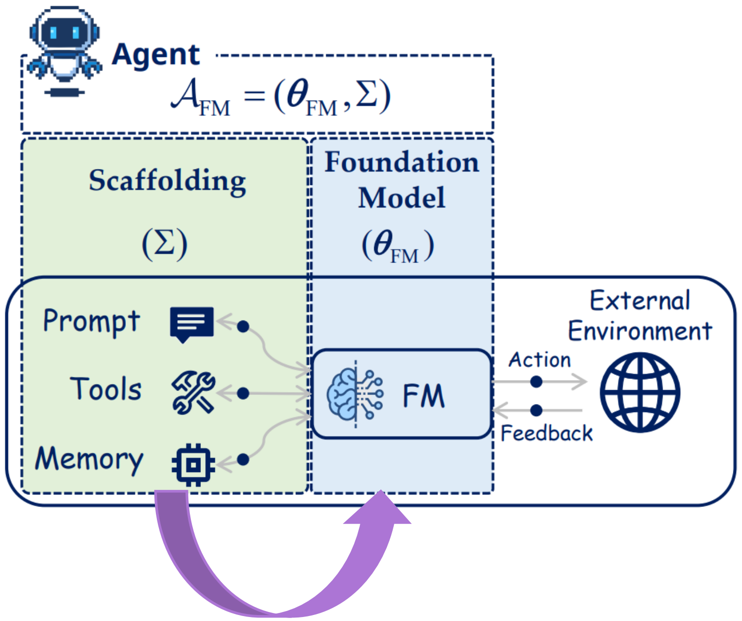
Survey on Self-Evolving Agents

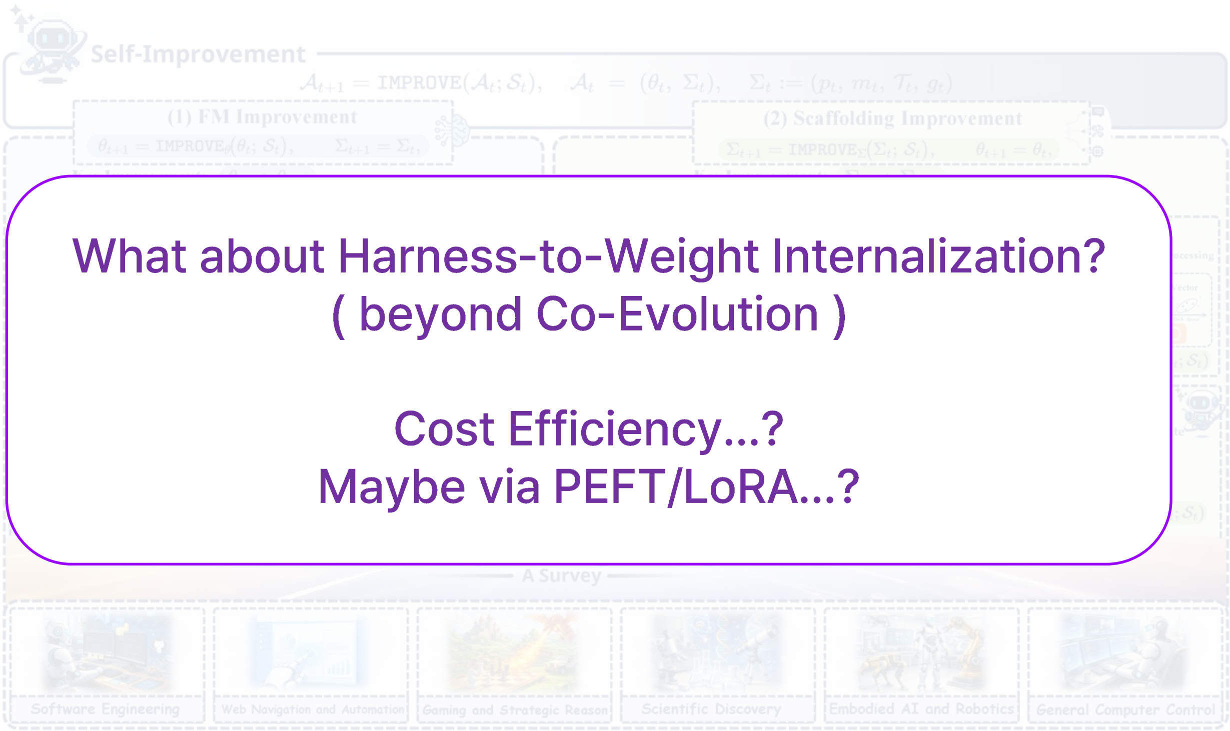
Model-Harness Co-Evolution; Using Data from Best Harness or Best Model



Initiating Questions

Harness-to-Weight Internalization via PEFT-RL





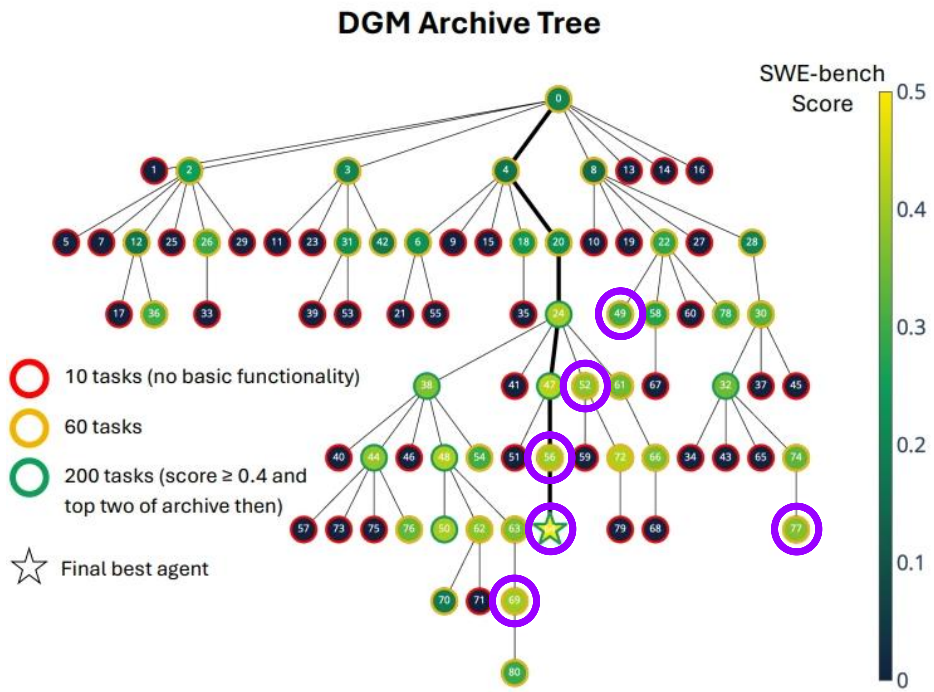
What about Harness-to-Weight Internalization?
(beyond Co-Evolution)

Cost Efficiency...?
Maybe via PEFT/LoRA...?

https://github.com/selfimproving-agent/Awesome-Self-Improving-Agents

Initiating Questions

Reusing Intermediate Harness to Generate Diverse Rollouts for RLVR



E.1 COST ESTIMATE

The estimated cost of completing a single run of the DGM on SWE-bench, as presented in Section 4, is about USD 22,000. In comparison, the estimated cost of completing a single run of either baseline (DGM w/o self-improve or DGM w/o open-ended exploration) on SWE-bench is about USD 10,000.

What about Reusing Intermediate Harness
for RL Environment Population?
(e.g., for diverse rollout generation)

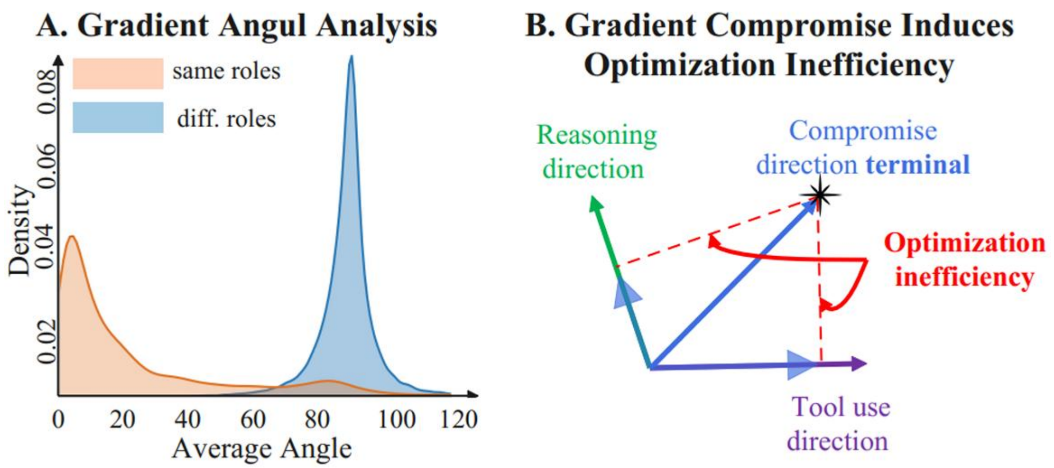
Just Wasted Harness...?
Reusing for RL Rather than SFT Distillation...?

cf. "Diverse" here means diversity at the high-level solution dimension or high reward variance, rather than token-level entropy.

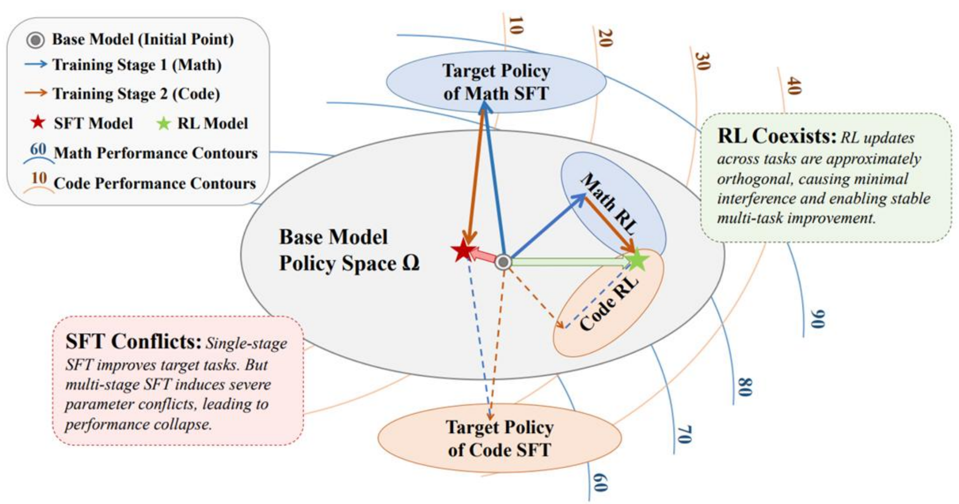
Research Questions

RQ1. How does harness self-evolution reshape the learning dynamics of Agentic RLVR?
Is the current best-performing harness also optimal for policy learning?

RQ2. How does harness-conditioned RLVR shape the parameter-update geometry?
Can LoRA initialization strategy be designed to better internalize these learning signals?



Gradient Misalignment between Two Token Roles (Reasoning & Tool-Use)



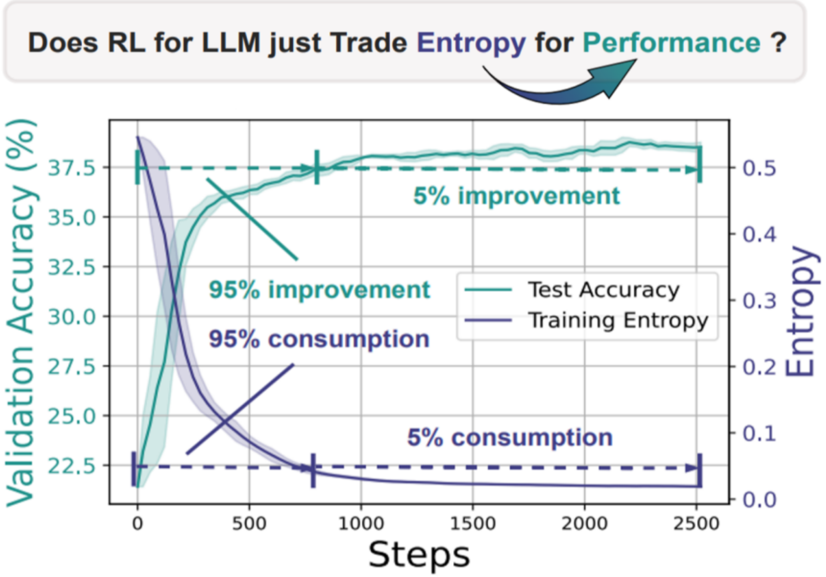
Orthogonal RL Parameter Updates across Different Tasks (esp. Multi-Stage RL)

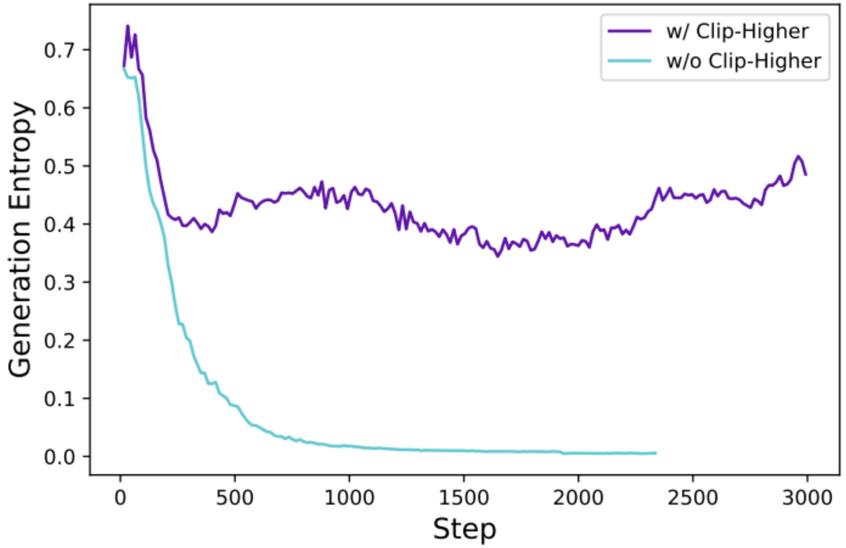
Reasoning and Tool-use Compete in Agentic RL:From Quantifying Interference to Disentangled Tuning: https://arxiv.org/abs/2602.00994
SFT Conflicts, RL Coexists: A Theoretical and Empirical Analysis of Multi-Task Learning for LLMs: https://arxiv.org/abs/2608.03573

Q&A

Why Rollout Diversity Matters

Entropy Collapse Problem in RLVR



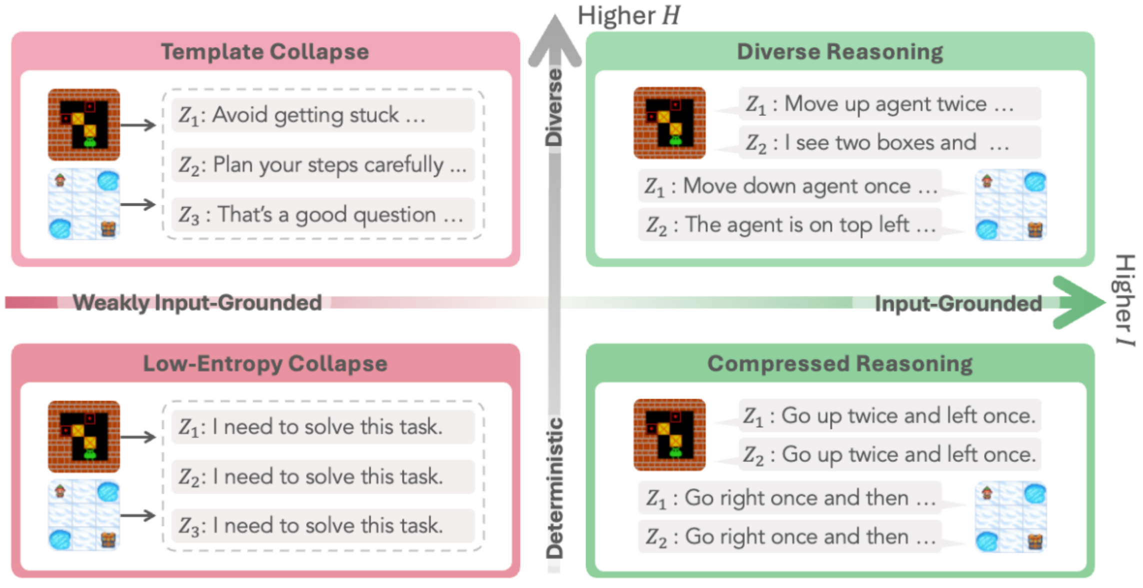


Although DAPO's Clip Higher helps mitigate entropy collapse,
it only alleviates the issue at the token level.

The Entropy Mechanism of Reinforcement Learning for Reasoning Language Models: https://arxiv.org/abs/2505.22617
DAPO: An Open-Source LLM Reinforcement Learning System at Scale: https://arxiv.org/abs/2503.14476

Why Rollout Diversity Matters

Entropy Collapse Problem in Agentic RLVR



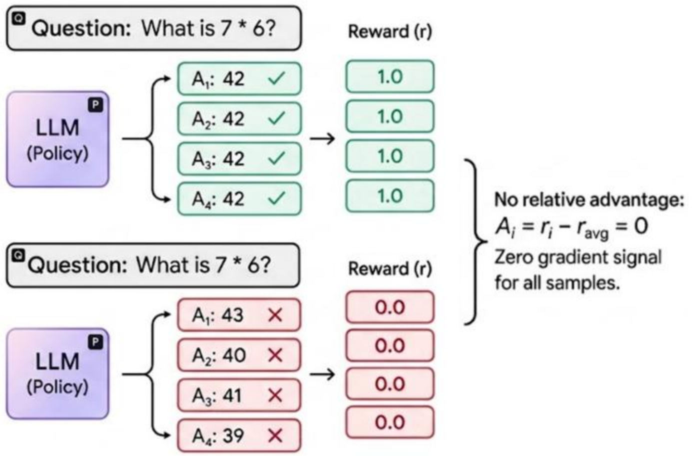
Four Quadrants of Reasoning on Entropy H and MI I

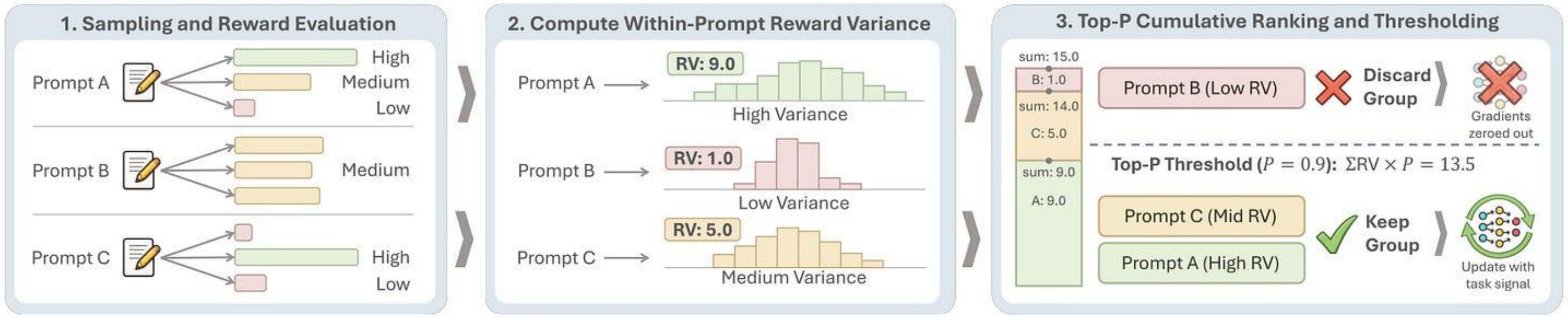
In RAGEN, it is highlighted that the entropy collapse problem in Agentic RLVR leads to behavior collapse and template collapse.

RAGEN-2: Reasoning Collapse in Agentic RL: https://arxiv.org/abs/2604.06268

Why Rollout Diversity Matters

Importance of Reward Distribution in GRPO Group



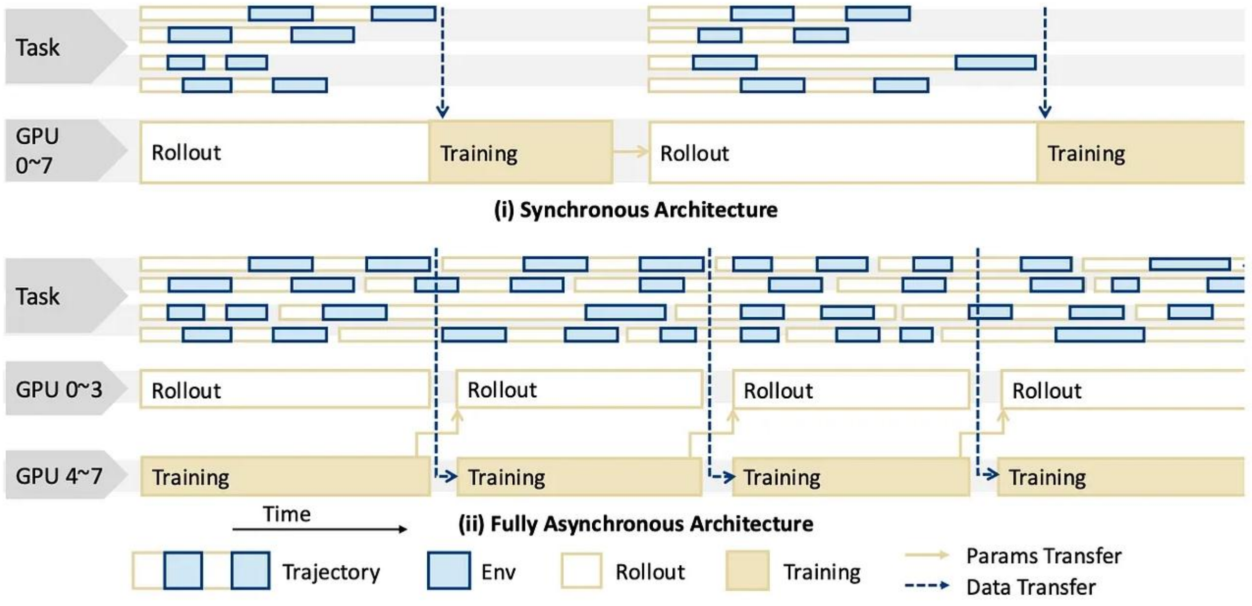


Due to the nature of GRPO, a learnable group cannot be formed if all rewards are identical; therefore, high reward variation is crucial for effective RLVR training.

RAGEN-2: Reasoning Collapse in Agentic RL: https://arxiv.org/abs/2604.06268

More about Agentic RLVR

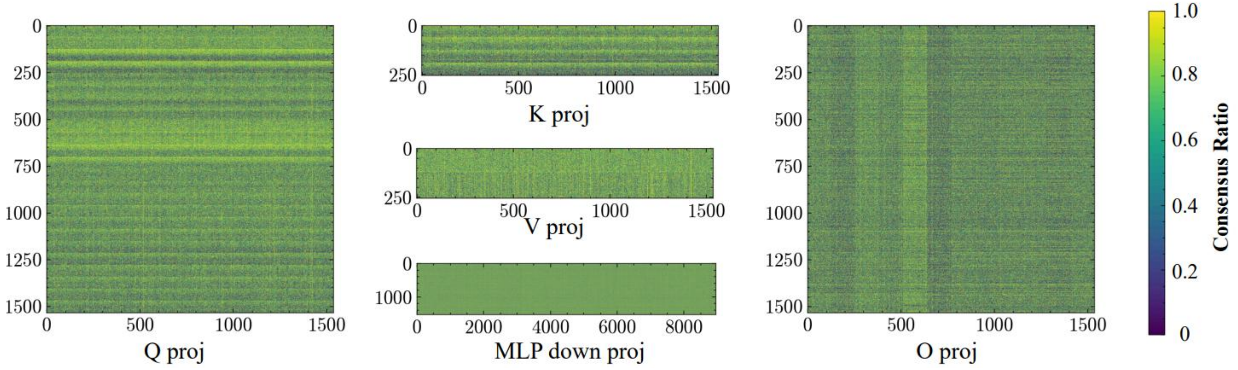
Asynchronous Rollout-Training System; Off-Policy Drifts

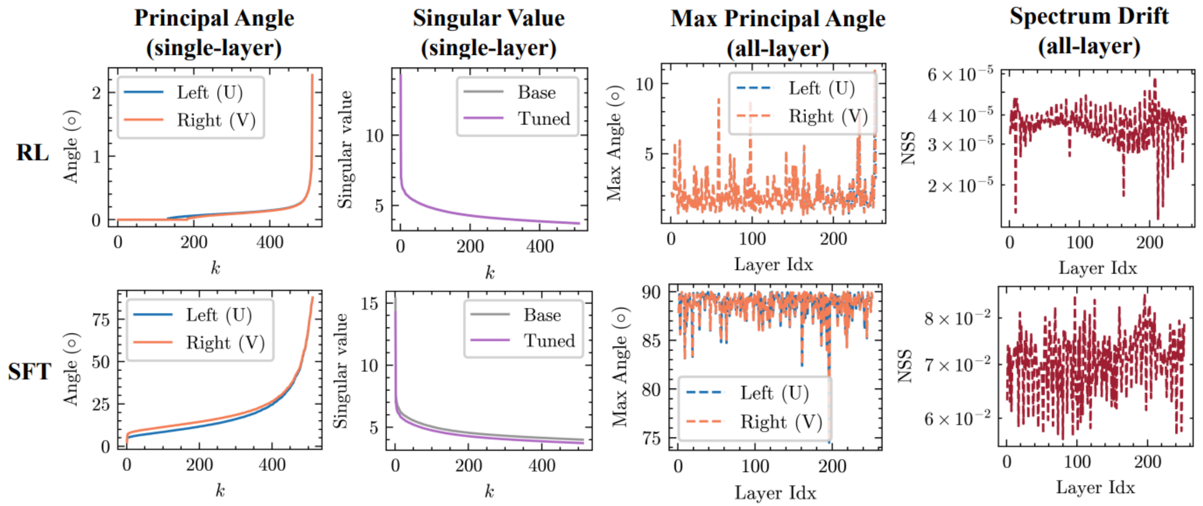


Since Agentic RLVR typically employs an asynchronous rollout-training system,
it is crucial to account for off-policy drift and staleness

Parameter-Level RLVR Training Dynamics

RLVR Provably Learns Off the Principals

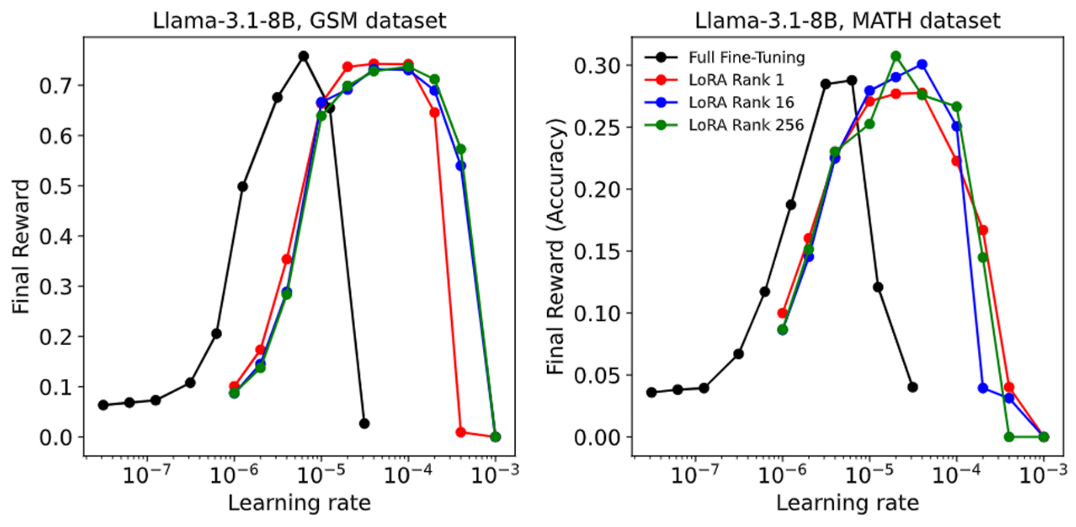


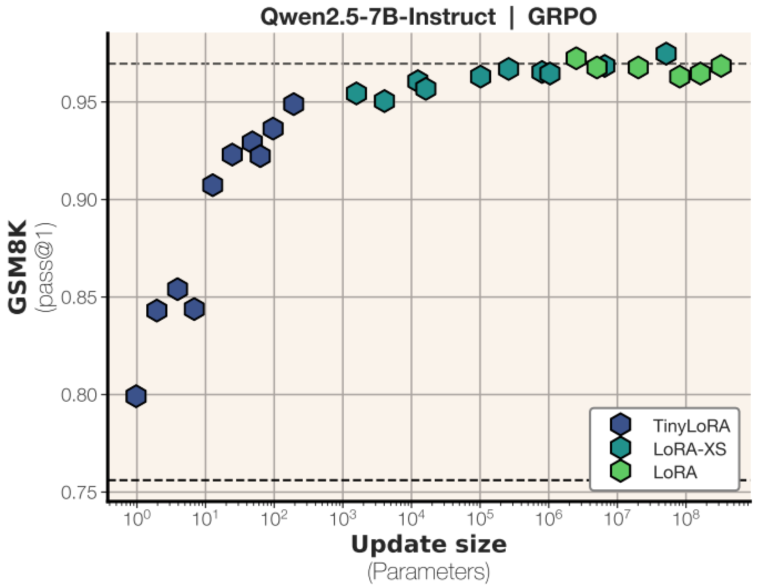


RLVR updates occur sparsely and in off-principal directions along the geometry of pre-trained model parameters

Parameter-Level RLVR Training Dynamics

PEFT LoRA for Reasoning





Emerging studies suggest that the sparse, off-principal updates in RLVR possess limited information density, making them highly amenable to low-rank LoRA approximation.

LoRA Without Regret: https://thinkingmachines.ai/blog/lora/
Learning to Reason in 13 Parameters: https://arxiv.org/abs/2602.04118

Revisiting PEFT-RL (esp. PEFT for RLVR)

Evaluating Parameter Efficient Methods for RLVR (25.12)

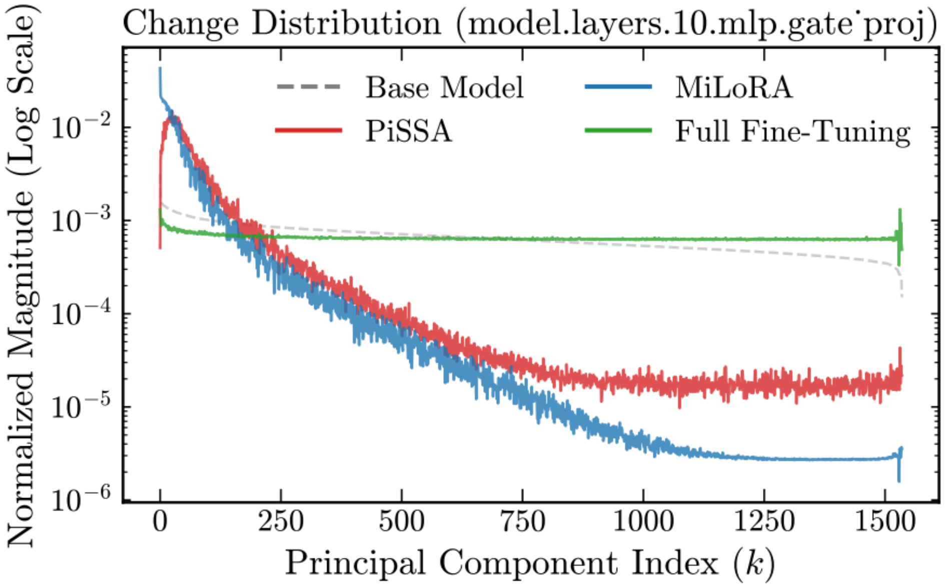
Methods	Avg.	AIME24@32		AIME25@32		AMC@32		HMMT@32		MATH500@4		Minerva@4	
		Avg.	Pass	Avg.	Pass	Avg.	Pass	Avg.	Pass	Avg.	Pass	Avg.	Pass
Baseline													
Base	40.5	32.4	76.7	22.2	33.3	69.0	90.0	9.9	23.3	72.0	86.8	14.2	25.0
Full	44.9	34.9	56.7	23.8	46.7	68.8	92.5	13.5	40.0	74.8	88.6	17.0	26.5
LoRA	42.5	33.2	60.0	22.9	36.7	64.4	95.0	13.3	33.3	72.1	87.4	13.5	23.5
Structural													
AdaLoRA	44.2	29.4	53.3	26.7	50.0	68.9	95.0	14.1	40.0	73.8	88.0	15.5	27.5
DoRA	46.6	39.0	80.0	28.8	43.3	71.9	95.0	13.4	36.7	75.8	90.0	14.7	26.5
MiSS	43.4	27.5	50.0	23.3	26.7	68.6	95.0	15.8	33.3	72.4	90.4	16.5	30.1
Efficient													
LoRA-FA	43.0	29.2	53.3	22.7	46.7	65.0	95.0	15.1	36.7	73.5	87.2	15.6	26.8
VeRA	40.7	29.1	60.0	21.7	36.7	61.5	95.0	14.4	30.0	69.7	85.6	13.1	24.6
Initialization													
LoRA+	43.9	28.1	50.0	25.9	50.0	70.0	95.0	16.5	36.7	72.2	90.4	15.4	26.5
rsLoRA	42.3	29.2	43.3	23.3	30.0	63.4	92.5	14.4	36.7	72.6	87.6	14.4	27.2
MiLoRA	18.0	4.2	6.7	0.0	0.0	19.6	47.5	0.0	0.0	44.5	63.4	11.7	19.9
PiSSA	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	1.0	0.1	0.4
Other PEFTs													
IA^3	22.3	0.7	6.7	3.5	10.0	25.2	57.5	0.6	3.3	55.4	71.4	12.8	22.1
LN Tuning	41.8	31.7	56.7	23.8	26.7	65.1	95.0	11.8	23.3	69.9	85.0	14.1	25.0

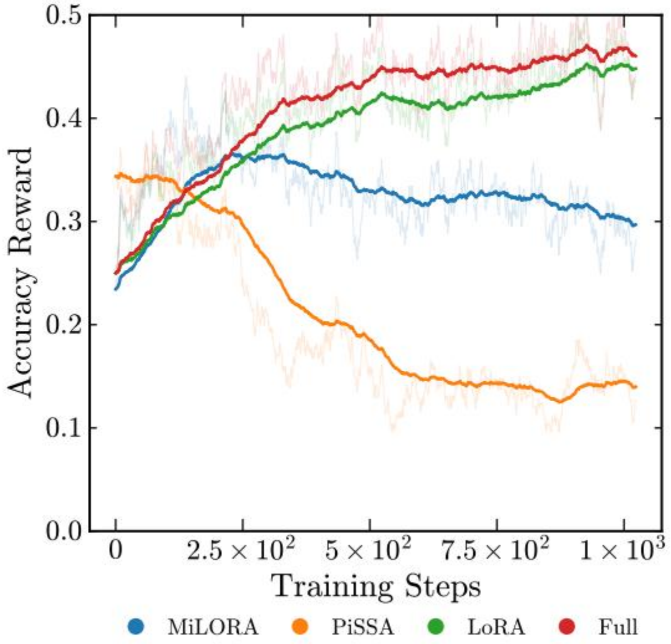
Table 3. Comparison of accuracy and pass scores. All values are reported in percentages.

Evaluating Parameter Efficient Methods for RLVR: https://arxiv.org/abs/2512.23165

Revisiting PEFT-RL (esp. PEFT for RLVR)

Evaluating Parameter Efficient Methods for RLVR (25.12)

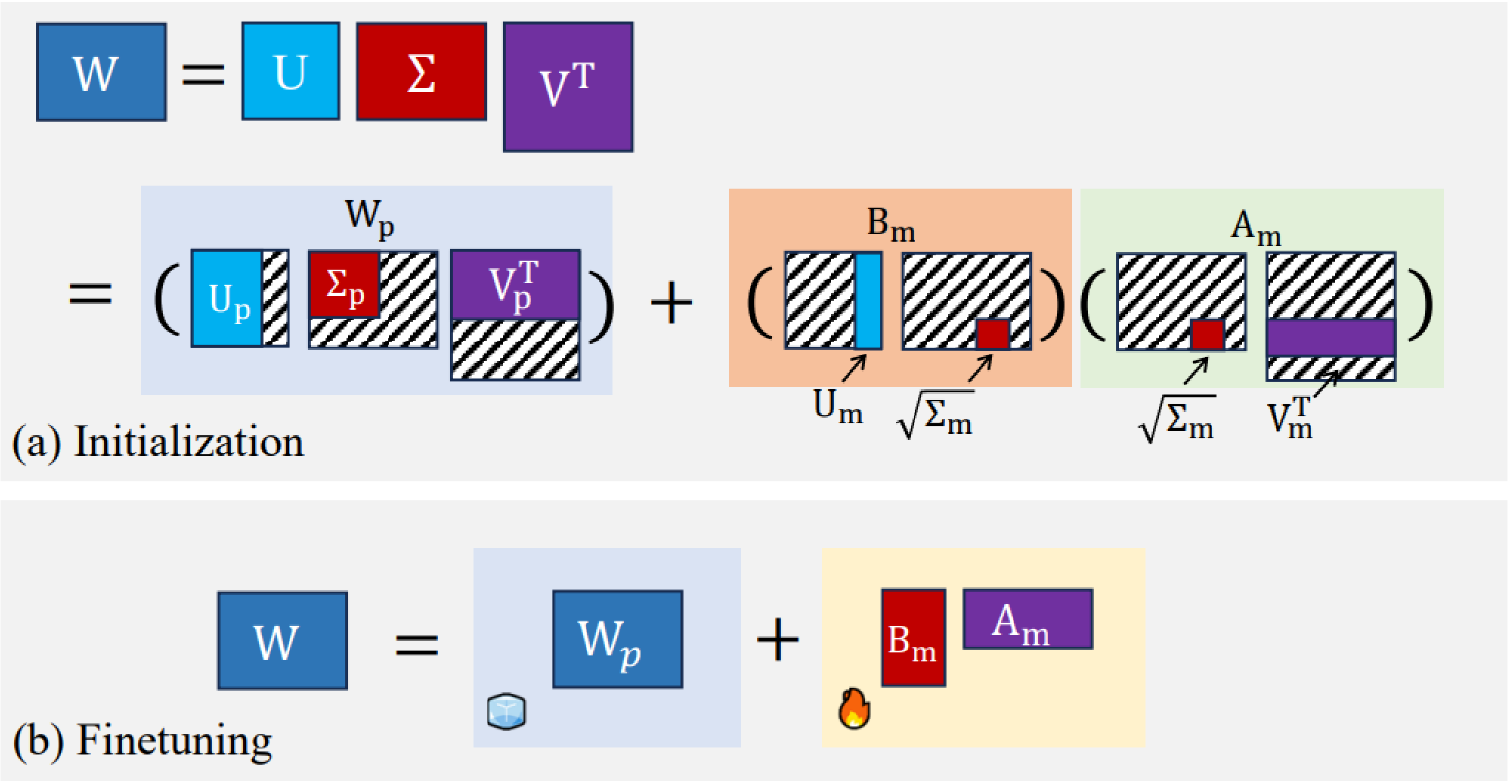




Finding 3: Existing SVD-based initialization strategies are unsuitable for RLVR. PiSSA fails due to a design conflict (forcing principal updates), creating a structural mismatch with RLVR's intrinsic tendency to **learn off the principals** (Zhu et al., 2025), resulting in training collapse. However, we find that MiLoRA fails due to optimization instability—negligible initialization magnitude causes the model to **degenerate back into principal-component updates**, failing to maintain the necessary off-principal trajectory.

High-Level Ideation

Off-Principal Targeting LoRA Initialization: Revisiting MiLoRA



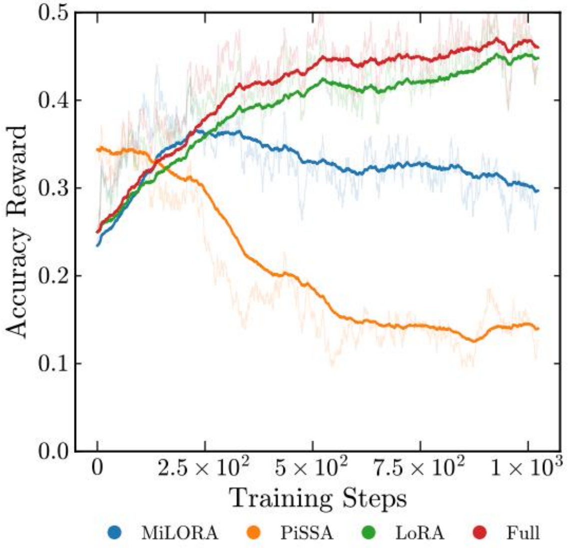
High-Level Ideation

Off-Principal Targeting LoRA Initialization: Revisiting MiLoRA

Despite the theoretical alignment, MiLoRA fails due to the disparity between initialization and gradient flow. We formalize the update dynamics at step t as:

$$\Delta \boldsymbol{W}_{t+1} \leftarrow \Delta \boldsymbol{W}_t - \eta \nabla \mathcal{L}(\boldsymbol{W}_t),\ \|\Delta \boldsymbol{W}_0\|_F = \|\boldsymbol{B}_{init} \boldsymbol{A}_{init}\|_F \approx 0 \text{ when } t = 0. \quad (3)$$

The minor singular values used for initialization satisfy $\sigma_{tail} \to 0$. Consequently, the initial adapter state effectively collapses to zero. This renders the intended structural constraint numerically non-existent. Without a significant initial bias $\|\Delta \boldsymbol{W}_0\|_F$, the optimization trajectory is dictated by the spectral properties of the gradient $\nabla \mathcal{L}$. Since the gradient aligns with the directions of maximum variance (principal components $\boldsymbol{U}_{:k}$), the update projects onto the principal subspace where $\langle \nabla \mathcal{L}, \boldsymbol{U}_{:k} \rangle \gg \langle \nabla \mathcal{L}, \boldsymbol{U}_{k:} \rangle$. Consequently, the updates are immediately reoriented by the dominant principal gradients, causing the model to degenerate from the off-principal regime back to the principal subspace, as evidenced by the spectral spike in Figure 3.



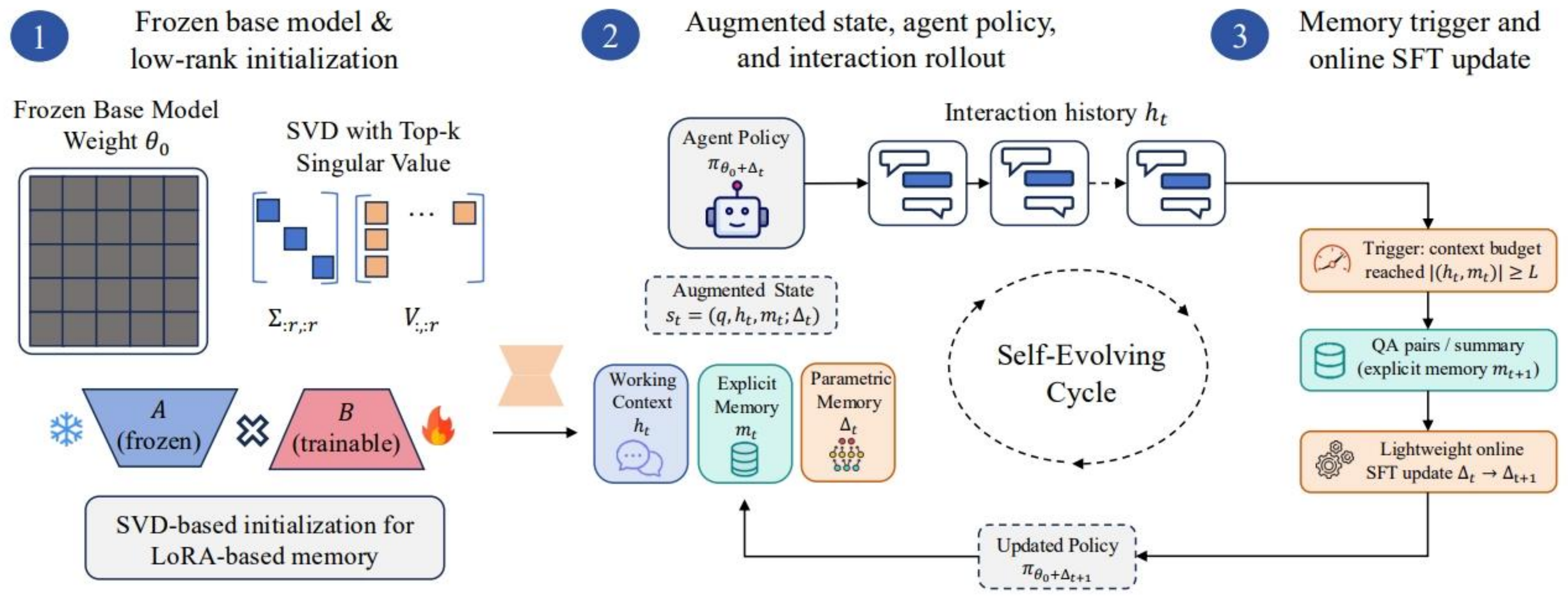
No Ablations for Higher Off-Principal Ranks;
What If More Trainable Parameters for Off-Principal Directions?

Or What about Scaling Singular Values? – Mathematical Verification Needed

Evaluating Parameter Efficient Methods for RLVR: https://arxiv.org/abs/2512.23165

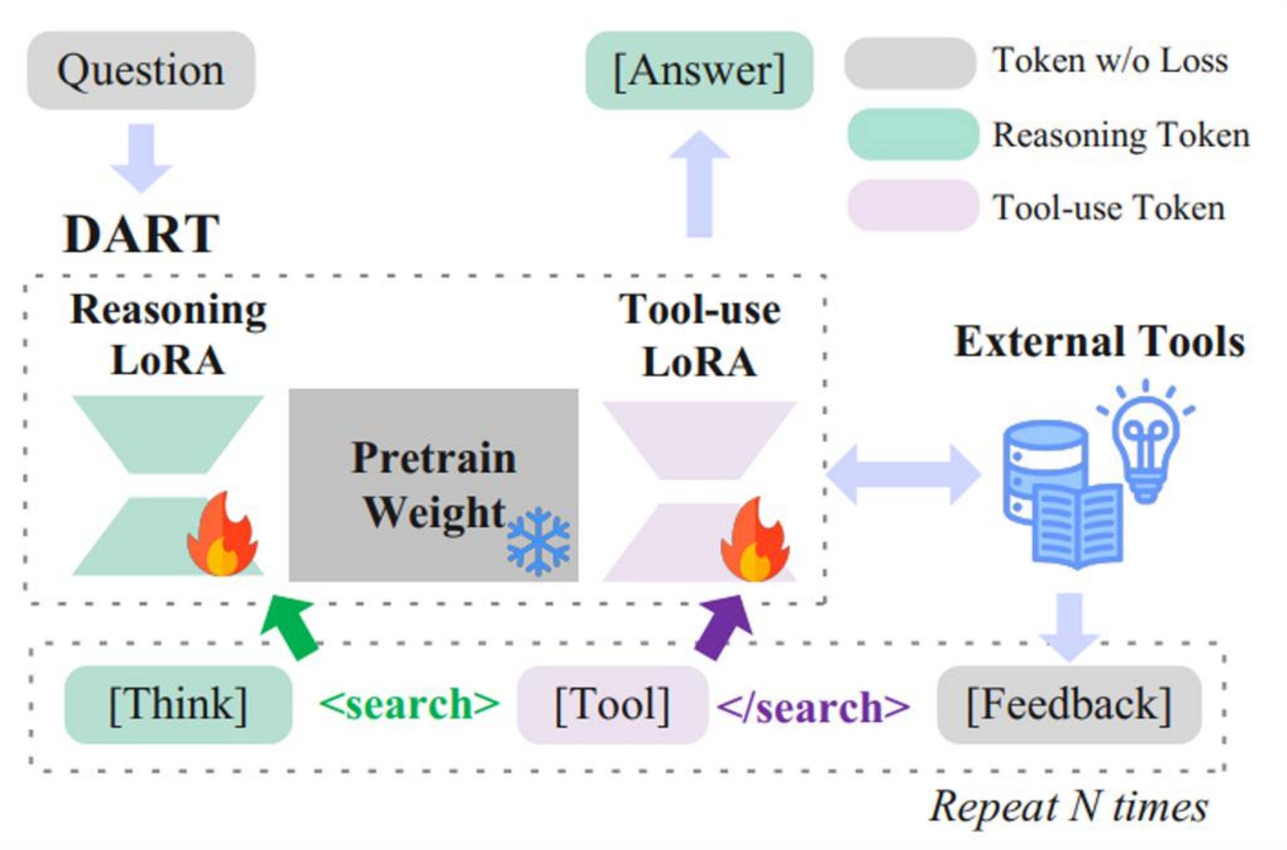
Extra Survey. Agentic PEFT

Intra-Episode Parametric Self-Evolution



Extra Survey. Agentic PEFT

Token Role-Specific LoRA Updates



Reasoning and Tool-use Compete in Agentic RL: From Quantifying Interference to Disentangled Tuning: https://arxiv.org/abs/2602.00994